Supplementary Analysis Appendix

Parallel elevation filtering of Ericaceae root-associated fungal communities in Andean páramo ecosystems

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Overview and reproducibility

This document compiles methodological details, supplementary analyses, and reproducibility outputs supporting the main manuscript. Computationally intensive steps (e.g. phylogenetic tree inference, GLLVM fitting, iNEXT3D diversity estimation) were executed on a remote Ubuntu-based computing environment. Their outputs (figures, tables, and selected model objects) are integrated here as static results.

All figures (PNG) and result tables (CSV) are stored within the project repository using relative paths. Rendering this document therefore requires cloning the repository and opening the .qmd file from the repository root.

Data availability.

Repository: Root_fungi_DADA2
Sequence data: ENA accession number PRJEB107725

Metadata: `Data_S1_metadata.csv, Data_S3_soil.csv`

If viewing the PDF version of this Appendix material, note that the HTML version has all code to reproduce these results accessible through drop-downs.

Please visit the project webpage for this: https://ssangulo.github.io/Paramo_SkyIslands_RootFungi/appendix.html

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Section 0: Setup

Output directory paths and shared aesthetic constants used throughout this document. All packages are loaded locally within each code section.

Section 1: Data and object loading

All pre-computed R objects are loaded here. Computationally intensive analyses (GLLVM, iNEXT3D, NTI/NRI, phylogeny inference) were executed on a remote server; their outputs are stored in `objects/` and loaded as needed.

Section 2: Bioinformatics summary and study design

2.1 Study sites and sampling

This study was conducted across four humid páramo regions in the Colombian Andes, spanning the Central and Eastern Cordilleras. Sampling locations covered elevations from 2,715 to 3,828 m a.s.l., representing the ecological transition from Andean forest to páramo (see main text, Fig. 1). The full sample metadata table is provided in Data_S1_metadata.csv.

Root sampling followed standardized sterile procedures. Fine roots of Gaultheria myrsinoides were excavated using a spade, and 3–5 cm root tips were clipped using a sterilized pruner. Tools were disinfected with 75% ethanol between samples, and gloves were changed between individual plants to minimize contamination. Root fragments were placed into sealed polybags stored in ice-cooled containers in the field. Samples were processed at Universidad ICESI, where roots were rinsed in sterile water to remove debris, dried briefly under a laminar flow hood, and stored at −20°C prior to shipment to Norway, and long-term storage at −80°C. Soil samples (top 15 cm) were collected at each sampling location, and five aliquots of rinse water were retained as laboratory and transport controls.

Table S1: GPS coordinates, elevation, and vegetation type classification of the 12 sampling locations across four páramo regions. These sites span the forest–subpáramo–páramo gradient and were used for root and soil sampling. In each location, 10 root samples were collected from 5 individuals (2 root samples per individual) and 1 soil sample.

Site ID	Site	Altitude (m)	Latitude (N)	Longitude (W)	Ecosystem
NV_1	Páramo La Nevera	3828	03º30’58.1”	076º03’05.4”	Páramo
NV_2	Páramo La Nevera	3550	03º31’37.7”	076º03’38.7”	Subpáramo
NV_3	Páramo La Nevera	3333	03º32’07.0”	076º04’18.3”	Andean forest
NV_4	Páramo La Nevera	3078	03º32’48.9”	076º04’31.3”	Andean forest
DOM_1	Páramo Las Domínguez	3815	03º43’51.8”	076º06’38.1”	Páramo

DOM_2	Páramo Las Domínguez	3534	03º44’07.2”	076º05’53.2”	Subpáramo
DOM_3	Páramo Las Domínguez	3234	03º44’17.1”	076º05’32.9”	Andean forest
BEL_1	Páramo Belmira	3254	06º38’43.8”	075º40’13.3”	Subpáramo
BEL_2	Páramo Belmira	2950	06º38’22.7”	075º39’57.3”	Andean forest
BEL_3	Páramo Belmira	2715	06°36’55.3”	075°39’51.1”	Andean forest
MA_1	Páramo Matarredonda	3678	04º33’31.5”	074º01’43.8”	Páramo
MA_2	Páramo Matarredonda	3381	04º33’10.3”	073º59’35.3”	Subpáramo

2.2 Read filtering, OTU table, and taxonomic assignment

This section provides supplementary details on sequencing read processing and quality filtering. The full bioinformatic pipeline (including DADA2 processing and replicate filtering) is available in the repository at “/Scripts/full_pipeline_DADA2.R”. Here we report summary tables of read retention and replicate-filtering thresholds used for downstream analyses.

Table S2: Median read numbers per PCR replicate for environmental samples (root and soil samples) and controls (blank extractions, positive, PCR blank, and tag-jump controls). The read counts are grouped at each stage of the DADA2 pipeline. Blank extraction controls (not containing any biological sample) were performed alongside real sample DNA extractions per each extraction batch. PCR blank controls contained all PCR reagents but no DNA template. Tag-jump controls are PCR reactions containing no primers and no DNA template, used to detect tag-jumping (index hopping).

	Raw reads	**Filtered**	**Merged - Non pooled**	**Merged - Pooled**	**Merged - Pseudopooled**
Root samples	77,047	71,783	71,084	70,343	71,242
Soil samples	72,158	66,555	65,311	65,498	65,385
Blank Extraction Controls	2,844	2,652	2,618	2,566	2,625
Positive Controls	95,238	88,174	88,062	87,996	87,972
PCR Blank Controls	291	261	250	257	256
Tag-jump Controls	41	36	27	30	32

Table 1: Table S3. Number of OTUs and samples after applying different replicate control filtering thresholds. The table shows results for both soil and root samples combined and only root samples.

Filtering_level	OTU_count_soil+root	Samples_soil+root	OTU_count_root_only	Samples_root_only
No filtering	18451	132	13225	120
OTU present in 2/4	7174	132	4386	120

Filtering_level	OTU_count_soil+root	Samples_soil+root	OTU_count_root_only	Samples_root_only
OTU present in 3/4	4870	131	2966	120
OTU present in 4/4	2875	113	1904	103

Table S4: Summary of the percentage of reads and OTUS with confident taxonomic assignment at each taxonomic rank, represented by habitat. Percentages were calculated from the filtered root-only dataset (soil reads removed).

Rank	**Habitat**	**%_reads_assigned**	**%_otus_assigned**
Phylum	Forest	88,1	88,1
Phylum	Subpáramo	91,6	88,6
Phylum	Páramo	92,9	89,3
Class	Forest	78,3	72,5
Class	Subpáramo	84,2	72,3
Class	Páramo	84,3	76,0
Order	Forest	68,3	59,2
Order	Subpáramo	74,2	57,1
Order	Páramo	78,2	60,9
Genus	Forest	32,8	34,2
Genus	Subpáramo	32,1	32,2
Genus	Páramo	31,4	32,7

2.3 Soil background subtraction

Soil controls were used as a third tier of negative control, alongside PCR blanks and extraction blanks, and were applied as a *count subtraction* rather than as a sample-level exclusion. Within each sampling location, the maximum read count observed per OTU across that location's co-sampled soil control(s) was subtracted from every root PCR replicate collected at the same location, and any resulting negative value was set to zero. The subtraction was applied to raw read counts, before the two-PCR-replicate detection filter and before PCR replicates were collapsed into samples; soil rows were then removed, leaving the root-only dataset used in all downstream analyses. If a location group contained no matching soil row the per-OTU maximum is undefined, and the affected values were reset to zero. The identical operation had already been applied one step earlier using PCR blanks and extraction blanks as the reference rows.

What this removes, and what it does not. Because the operation subtracts a background level rather than deleting shared taxa, it does not exclude fungi that occur in both soil and roots. An OTU is lost from a given root sample only where its root read count does not exceed the soil maximum at that location; taxa genuinely enriched in roots retain their excess signal. Consistent with this, 1,269 of the 3,689 OTUs recovered from soils were also retained in the final root dataset, and *Leohumicola* — abundant in both compartments — remains one of the dominant root genera (main-text Figure 4C). Root-associated fungi with a soil interface, and dormant propagules of root-associated taxa residing in the soil pool, are therefore represented in the analysed data. The rationale is not that soil-derived taxa are uninformative, but that each root sample is corrected against soil from its own location, which removes precisely the component of root community variation that tracks local soil composition; habitat signal that survives this correction is consequently not attributable to soil carry-over from incomplete root washing or recalcitrant surface-bound cells.

Depth dependence — an acknowledged limitation. The subtraction operates on raw counts and is not normalised for the sequencing depth of the soil control relative to each root sample. Median depths were comparable between the two sample types (65,311 versus 71,084 reads per PCR replicate after merging under the non-pooled strategy; Table S2), so the correction is not grossly asymmetric, but it remains depth-dependent: a fixed count is removed from root samples that differ in depth, so its proportional effect is larger in shallow-sequenced samples and largest for taxa spanning the soil–root continuum. The procedure therefore trades one source of bias (incomplete root washing, surface-bound soil taxa) for another (depth-dependent magnitude of count removal), and it is conservative with respect to root–soil overlap: genuine overlap is removed together with carry-over. The communities analysed here are accordingly best interpreted as the root-enriched fraction of the fungal community rather than as the complete root mycobiome.

Section 3: Taxonomic overview

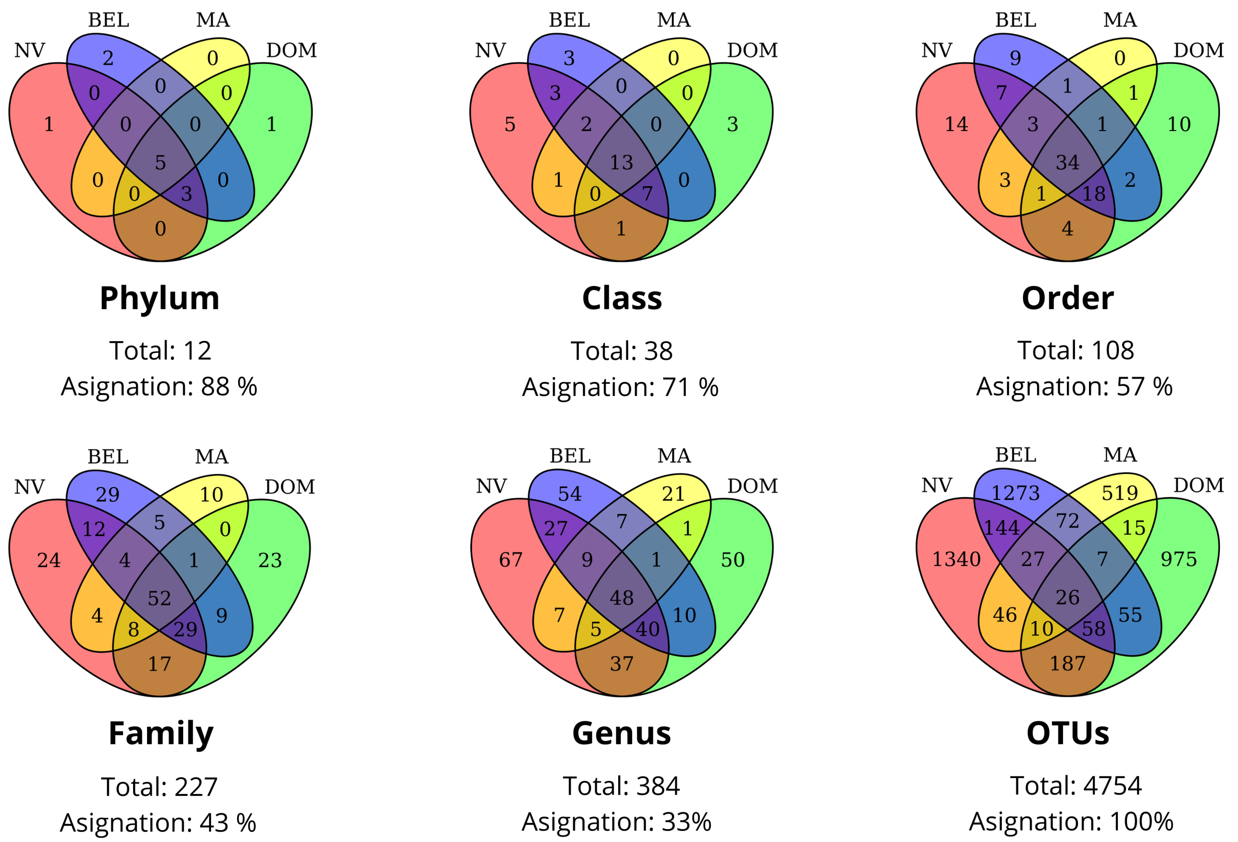

Figure S1: Venn diagram of taxonomic assignments across sampling sites, in the root only dataset, but previous to soil sequences removal. Venn diagrams illustrate the overlap of OTU taxonomic assignments across four study sites: La Nevera (NV), Las Domínguez (DOM), Belmira (BEL), and Matarredonda (MA). Only OTUs with confirmed assignments at each taxonomic level are included.

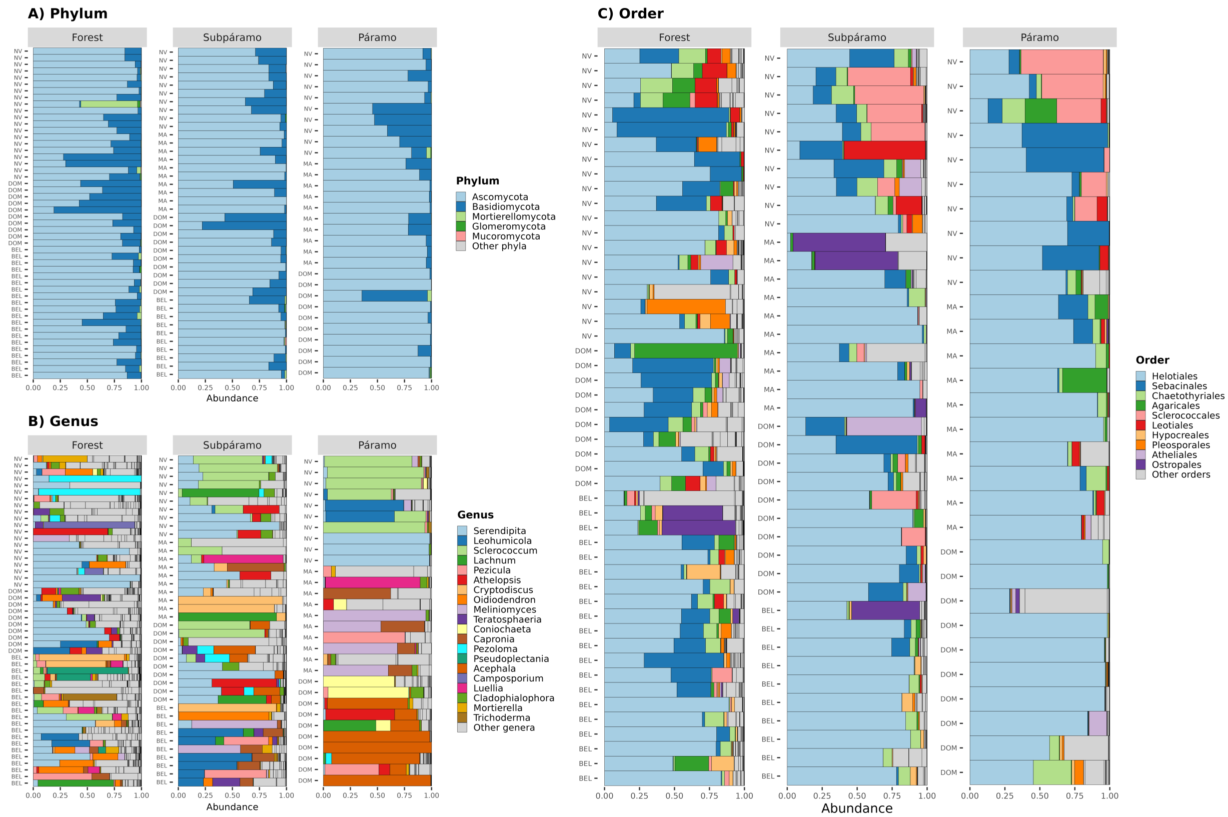

Figure S2: Taxonomic composition in root samples along the elevational gradient. Relative read abundances are shown at three taxonomic ranks: A) Phylum, B) Genus and C) Order. Each bar represents an individual root sample, grouped by habitat: Forest (2700-3300 m), Subpáramo (3300-3500 m), and Páramo (3800 m). Study sites are indicated on the y-axis.

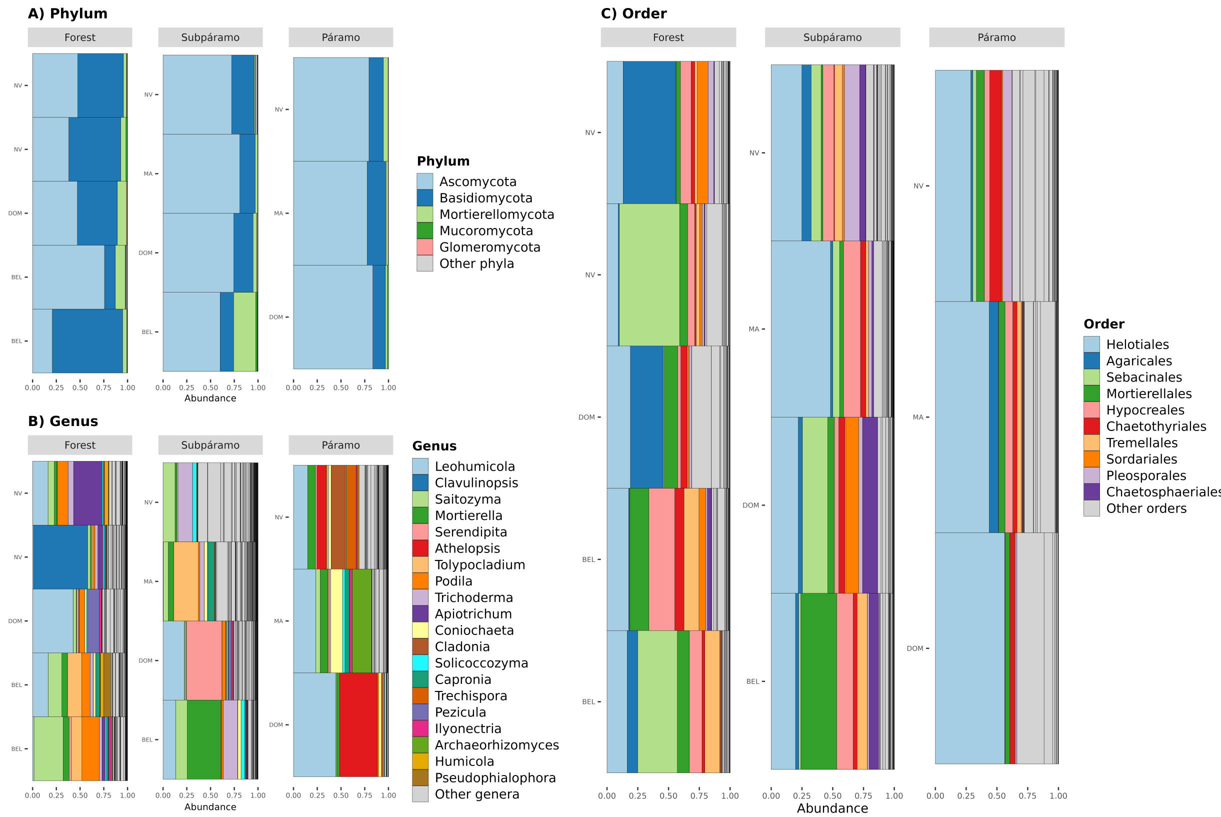

Figure S3: Taxonomic composition in soil samples, collected at each of the sampling locations. Relative read abundances are shown at three taxonomic ranks: A) Phylum, B) Genus, and C) Order. Each bar represents an individual soil sample, grouped by habitat: Forest (2700-3300 m), Subpáramo (3300-3500 m), and Páramo (3800 m). Study sites are indicated on the y-axis.

Section 4: Community composition

This section provides detailed statistical analyses supporting the patterns of root-associated fungal community composition described in the main text. Several of these analyses form the basis of results reported in the main manuscript; here we document the full model specifications and test statistics to ensure transparency and reproducibility.

We include distance-based redundancy analysis (dbRDA) based on robust Aitchison distances to assess habitat-associated compositional structure while conditioning on site. We further present PERMANOVA results from a balanced subset of samples (NV–DOM) used to test habitat and site effects under a controlled sampling design. In addition, we report a centroid-based turnover model quantifying the magnitude of community change across habitats using the full dataset.

4.1 dbRDA ordination (Tables S5–S6)

Distance-based redundancy analysis (dbRDA) was conducted on robust Aitchison distances to test for habitat-associated compositional structure while conditioning on site. This analysis is reported in the main manuscript as Figure 2A (habitat separation conditioned on site; permutation test $F = 1.41$, $p = 0.001$).

Table 2: Table S5. dbRDA permutation tests (by term; 999 permutations).

Component	Df	Variance	F	Pr
habitat	2	22.718	1.408	0.001
Residual	54	435.501	NA	NA

Table 3: Table S6. dbRDA permutation tests (by constrained axis; 999 permutations).

Axis	Df	Variance	F	Pr
CAP1	1	14.579	1.808	0.001
CAP2	1	8.139	1.009	0.470
Residual	54	435.501	NA	NA

4.2 PERMANOVA and beta-dispersion (Tables S7–S9)

Beta-dispersion tests indicated higher within-habitat variability in forests compared with subpáramo and páramo communities ($p < 0.001$), whereas dispersion did not differ among sites ($p = 0.526$). Because the PERMANOVA was conducted on a balanced subset of samples, the significant effects of habitat, site, and their interaction are interpreted as reflecting genuine compositional differences rather than artifacts of unequal dispersion.

Table 4: Table S7. PERMANOVA (adonis2) results by term.

Source	df	SumOfSqs	R2	Pseudo.F	Pr..F.
site	1	742.078	0.056	1.786	0.001
habitat	2	1351.164	0.102	1.626	0.001
site:habitat	2	1241.269	0.093	1.493	0.001
Residual	24	9974.546	0.749	NA	NA
Total	29	13309.057	1.000	NA	NA

Table 5: Table S8. Habitat beta-dispersion (distance to centroid). Permutation test p = 0.001.

habitat	avg_distance_to_centroid
forest	23.631
paramo	17.147
subparamo	17.794

Table 6: Table S9. Site beta-dispersion permutation test.

test	permutations	F	p_value
betadisper_site	999	0.314	0.571

4.3 Centroid-based turnover model (Table S10)

To quantify the magnitude of community turnover across habitats while accounting for spatial structure, we calculated the Aitchison distance of each sample to its site-specific centroid and modelled turnover as a function of habitat, site, and their interaction. Habitat had a strong effect on turnover magnitude, whereas the habitat × site interaction was not significant, indicating consistent habitat-associated turnover across geographically isolated páramo regions (Table S10).

Panel B in the main manuscript Figure 2 summarizes this model (habitat $F_{2,50} = 14.7$, $p < 0.001$; habitat × site $F_{4,50} = 1.28$, $p = 0.29$), with Matarredonda (MA) shown as a dashed trend due to a compressed gradient.

Table 7: Table S10. Linear model results for centroid-based turnover.

Effect	df	Sum.of.Squares	Mean.Square	F.value	p.value
habitat	2	376.745	188.373	14.719	0.000
site	3	138.096	46.032	3.597	0.020
habitat:site	4	65.496	16.374	1.279	0.291
Residuals	50	639.914	12.798	NA	NA

4.4 Figure 2

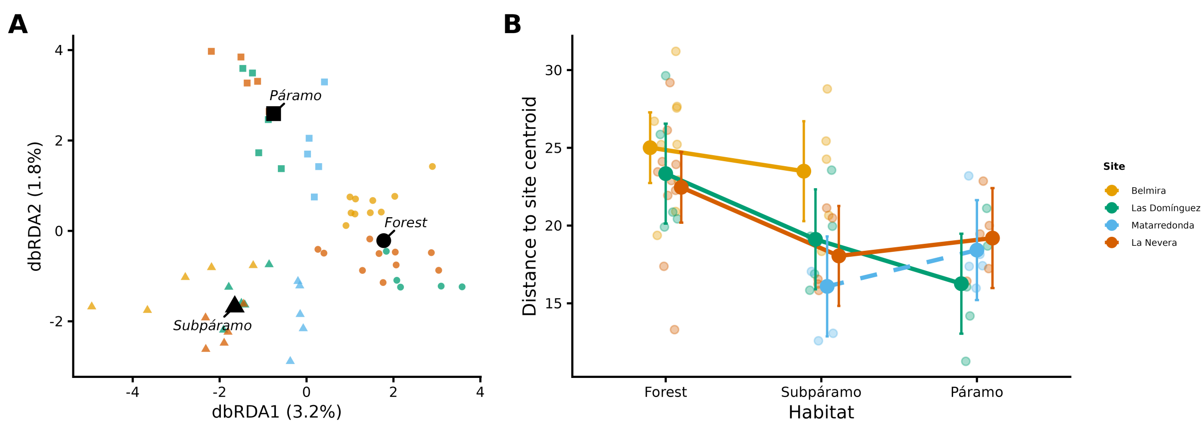

Figure 2: Community composition and turnover across habitats. Panel A: dbRDA ordination of RAF communities conditioned on site (color = site, shape = habitat; permutation $F = 1.41$, $p = 0.001$). Panel B: centroid-based turnover model (distance to site-specific centroid across habitats), with MA shown as dashed due to incomplete gradient (habitat $F_{2,50} = 14.7$, $p < 0.001$; habitat × site $F_{4,50} = 1.28$, $p = 0.29$).

Section 5: Diversity trajectories

Diversity trajectories were assessed using three complementary approaches: per-sample Hill diversity models to test parallelism across sites, iNEXT3D rarefaction/extrapolation curves for habitat-level estimates, and beta-diversity partitioning. All analyses require a rooted phylogenetic tree, inferred below. Analyses are presented in computation order; Figure 3 follows as a composite of pre-computed outputs.

Alignment and tree inference. OTU reference sequences (ITS2; the 4,386 OTUs of the replicate-filtered, soil-subtracted root dataset, retaining both root replicates per plant) were aligned with the profile-to-profile aligner `AlignSeqs` in DECIPHER (Wright 2016), with sequence anchoring disabled (`anchor = NA`). No post-alignment curation, masking or trimming was applied. ITS2 is length-variable and hypervariable, so automated trimming heuristics discard a large share of the phylogenetically informative sites; and because the alignment is used only to derive a single fixed topology on which all samples are subsequently compared, alignment uncertainty enters every sample's diversity estimate identically rather than differentially by habitat. A neighbour-joining tree built from maximum-likelihood distances (`dist.ml`) provided the starting topology, which was then optimised by maximum likelihood with `optim.pml` in phangorn (Schliep 2011) under GTR + Γ + I with four discrete rate categories, simultaneously optimising base frequencies, the rate matrix, branch lengths, the gamma shape parameter and the proportion of invariant sites, using stochastic rearrangements. The tree was inferred unrooted and midpoint-rooted (`phangorn::midpoint`) before use in phylogenetic diversity, NRI and NTI calculations.

Substitution model. GTR + Γ + I was adopted a priori rather than selected by a formal model-selection test, following the standard DADA2/phyloseq phylogenetic workflow (Callahan et al. 2016). It is the most parameter-rich time-reversible model available in `optim.pml`, so under-parameterisation is not a concern, and a formal model test across 4,386 ITS2 sequences was computationally prohibitive at this scale. All phylogenetic metrics reported here (Faith's PD, phylogenetic Hill numbers, NRI and NTI) are comparative statistics computed from relative branch lengths on this single shared topology, and are therefore expected to be robust to moderate differences in substitution model.

5.1 Abundance-based beta-diversity partitioning (Table S11)

Bray–Curtis dissimilarities were partitioned into balanced variation and abundance gradients across taxonomic levels using the `betapart` package. Metrics were calculated from rarefied abundance data.

Table 8: Table S11. Bray–Curtis beta-diversity partitioning across taxonomic levels and sites.

Site	Taxonomic_Level	beta.BRAY.BAL	beta.BRAY.GRA	beta.BRAY
DOM	OTU	0.9748	0.0244	0.9993
DOM	Genus	0.9164	0.0780	0.9944
DOM	Order	0.7708	0.2166	0.9874
DOM	Class	0.4256	0.5418	0.9674
DOM	Phylum	0.2759	0.6859	0.9619
NV	OTU	0.9870	0.0124	0.9995
NV	Genus	0.9553	0.0415	0.9969
NV	Order	0.7846	0.2059	0.9905
NV	Class	0.4516	0.5238	0.9754
NV	Phylum	0.1470	0.8046	0.9516
BEL	OTU	0.9870	0.0125	0.9994
BEL	Genus	0.9371	0.0573	0.9945
BEL	Order	0.7916	0.1995	0.9911
BEL	Class	0.4611	0.5121	0.9733
BEL	Phylum	0.0258	0.9434	0.9692
MA	OTU	0.9431	0.0557	0.9989
MA	Genus	0.9305	0.0627	0.9932
MA	Order	0.7966	0.1911	0.9877
MA	Class	0.5906	0.3750	0.9656
MA	Phylum	0.0000	0.9504	0.9505

5.2 iNEXT3D rarefaction curves and asymptotic diversity summaries (Tables S12–S13)

We quantified habitat-associated patterns in taxonomic diversity (TD) and phylogenetic diversity (PD) using iNEXT3D and Hill numbers (q = 0, 1, 2). Diversity estimates were computed from incidence (presence/absence) matrices at the habitat level and evaluated under rarefaction/extrapolation with bootstrap confidence intervals (nboot = 500). Phylogenetic diversity was estimated as meanPD using a pruned phylogeny matched to the observed taxa. The combined curve visualization is presented in main-text Figure 3A; asymptotic summaries are provided in Tables S12–S13, and the per-sample parallelism ANOVA results in Tables S14–S15.

Table 9: Table S12. Taxonomic diversity (TD) summary from iNEXT3D across habitats (q = 0, 1, 2; nboot = 500). Observed and asymptotic estimates are reported with standard errors, 95% confidence intervals, and sample coverage at observed effort and at double effort.

Assemblage	qTD	TD_obs	TD_asy	s.e.	qTD.LCL	qTD.UCL
Forest	Species richness	2521.000	4581.983	125.848	4335.326	4828.640
Forest	Shannon diversity	1810.823	2927.463	51.242	2827.030	3027.896
Forest	Simpson diversity	1158.528	1467.726	34.168	1400.758	1534.695
Páramo	Species richness	901.000	1945.702	95.214	1759.086	2132.317
Páramo	Shannon diversity	693.768	1316.325	43.456	1231.153	1401.497
Páramo	Simpson diversity	474.571	661.005	30.584	601.061	720.948
Subpáramo	Species richness	1533.000	3059.820	108.142	2847.866	3271.775
Subpáramo	Shannon diversity	1225.295	2341.860	58.351	2227.494	2456.226
Subpáramo	Simpson diversity	846.718	1247.075	49.323	1150.403	1343.747

Table 10: Table S13. Phylogenetic diversity (PD; meanPD) summary from iNEXT3D across habitats (q = 0, 1, 2; nboot = 500), including coverage and effective lineage estimates under rarefaction/extrapolation.

Habitat	qPD	PD_obs	PD_asy	s.e.	qPD.LCL	qPD.UCL	Reftime	Type
Forest	q = 0 PD	185.553	283.343	7.643	268.362	298.324	0.873	meanPD
Forest	q = 1 PD	96.045	115.843	1.427	113.045	118.641	0.873	meanPD
Forest	q = 2 PD	53.644	56.697	0.585	55.551	57.842	0.873	meanPD
Páramo	q = 0 PD	81.971	134.076	6.501	121.334	146.818	0.873	meanPD
Páramo	q = 1 PD	46.984	60.900	1.321	58.310	63.489	0.873	meanPD
Páramo	q = 2 PD	27.101	29.356	0.529	28.320	30.392	0.873	meanPD
Subpáramo	q = 0 PD	132.936	213.016	7.909	197.514	228.518	0.873	meanPD
Subpáramo	q = 1 PD	75.157	98.360	1.843	94.748	101.973	0.873	meanPD
Subpáramo	q = 2 PD	39.842	43.233	0.715	41.831	44.635	0.873	meanPD

5.3 Per-sample Hill diversity: parallelism models (Tables S14–S15)

To evaluate whether diversity trajectories are parallel across geographically isolated sites, we model diversity as a function of habitat (ordered along elevation), site, and their interaction. The key term is habitat × site: a weak/non-significant interaction with a strong habitat main effect supports broadly parallel trajectories with site-specific baselines.

For this section, diversity is computed independently for each sample using Hill numbers: taxonomic diversity (TD) and phylogenetic diversity (PD) at q = 1 (effective Shannon) as the primary metric. This preserves interpretability, keeps consistency with Hill-number logic, and allows direct sample-level inference on trajectory parallelism.

We use a two-model strategy: (i) a primary habitat-factor model for ecological interpretation, and (ii) a secondary numeric-trend model (ordered habitat steps) to summarize site-specific slopes. Model family is screened first (Gaussian, log-Gaussian, Gamma-log) before final inference.

Post-hoc pairwise habitat contrasts (Table S15b). Habitat is a categorical predictor, so the omnibus F-test for the habitat term establishes that habitats differ but does not identify which pairs differ, or in which direction. We therefore followed it with pairwise contrasts among the three habitats. The habitat × site interaction is non-significant for both metrics (Tables S14–S15), and the full design contains empty habitat × site cells (Matarredonda has no forest, Belmira no páramo), which makes habitat marginal means non-estimable from the interaction model; pairwise contrasts were therefore estimated from the corresponding additive model, log(diversity) ~ habitat + site, and Tukey-adjusted for the three comparisons. Contrasts are reported on the response scale as ratios of effective diversity, so a ratio above one indicates higher diversity in the first habitat.

The direction of every contrast is consistent with a decline along the gradient: estimated marginal diversity falls from forest (TD 18.3, PD 1.96) to subpáramo (13.2, 1.70) and páramo (13.3, 1.66), and both forest-versus-higher-habitat ratios exceed one (TD forest/subpáramo 1.39, forest/páramo 1.38; PD 1.15 and 1.18). No individual pairwise contrast is resolved after correction for multiple comparisons, however (all Tukey-adjusted p 0.17; Table S15b), and the subpáramo–páramo contrast is essentially flat (TD ratio 0.99, PD 1.03). The significant omnibus habitat term for TD (Table S14) is thus carried by the combined forest-versus-páramo-belt difference rather than by any single resolved pair, and the post-hoc tests give no additional support for a stepwise decline across successive habitats at the per-sample level. Two features of the data limit resolution here: the per-habitat sample sizes available for pairwise inference are small (n = 25, 20 and 15 for forest, subpáramo and páramo), and the flat subpáramo–páramo contrast at the sample level contrasts with the habitat-level rarefaction curves, where páramo is clearly the least diverse assemblage (Figure 3A, Tables S12–S13). Taken together these suggest that the further decline into páramo detected at the habitat (gamma) level reflects reduced among-sample turnover rather than lower per-sample (alpha) diversity — a distinction the omnibus habitat test alone does not expose.

Table 11: Table S14. ANOVA table for the log-linear model of per-sample taxonomic diversity (TD; Hill q=1) as a function of habitat, site, and their interaction. Habitat $\times$ site interaction F(4,50)=1.52, p=0.21 supports parallel diversity trajectories across sites.

Term	Df	Sum.Sq	Mean.Sq	F.value	Pr..F.
habitat	2	2.748	1.374	4.395	0.017
site	3	0.633	0.211	0.675	0.571
habitat:site	4	1.899	0.475	1.518	0.211
Residuals	50	15.633	0.313	NA	NA

Table 12: Table S15. ANOVA table for the log-linear model of per-sample phylogenetic diversity (PD; Hill q=1, meanPD) as a function of habitat, site, and their interaction. Habitat $\times$ site interaction F(4,50)=1.22, p=0.31 supports parallel diversity trajectories across sites.

Term	Df	Sum.Sq	Mean.Sq	F.value	Pr..F.
habitat	2	1.135	0.568	3.027	0.057
site	3	0.794	0.265	1.411	0.251
habitat:site	4	0.918	0.230	1.224	0.313
Residuals	50	9.375	0.188	NA	NA

Table 13: Table S15b. Post-hoc pairwise contrasts among habitats for per-sample Hill diversity (q=1), for taxonomic (TD) and phylogenetic (PD) diversity. Contrasts were estimated from the additive models log(diversity) ~ habitat + site and Tukey-adjusted for three comparisons; the additive model is used because the habitat $\times$ site interaction is non-significant (Tables S14–S15) and the full design contains empty habitat $\times$ site cells, which makes habitat marginal means non-estimable from the interaction model. Estimates are back-transformed to the response scale: ratio is the multiplicative difference in effective diversity between the two habitats (ratio > 1 = higher diversity in the first habitat), with 95% confidence limits.

metric	contrast	ratio	SE	df	lower.CL	upper.CL	t.ratio	p.value
TD (q=1)	forest / subparamo	1.390	0.252	54	0.897	2.152	1.813	0.175
TD (q=1)	forest / paramo	1.382	0.291	54	0.832	2.297	1.536	0.282
TD (q=1)	subparamo / paramo	0.995	0.199	54	0.614	1.612	-0.027	1.000
PD (q=1)	forest / subparamo	1.152	0.160	54	0.824	1.611	1.021	0.567

metric	contrast	ratio	SE	df	lower.CL	upper.CL	t.ratio	p.value
PD (q=1)	forest / paramo	1.184	0.191	54	0.802	1.747	1.046	0.551
PD (q=1)	subparamo / paramo	1.027	0.158	54	0.710	1.487	0.176	0.983

5.4 Figure 3: Diversity trajectories

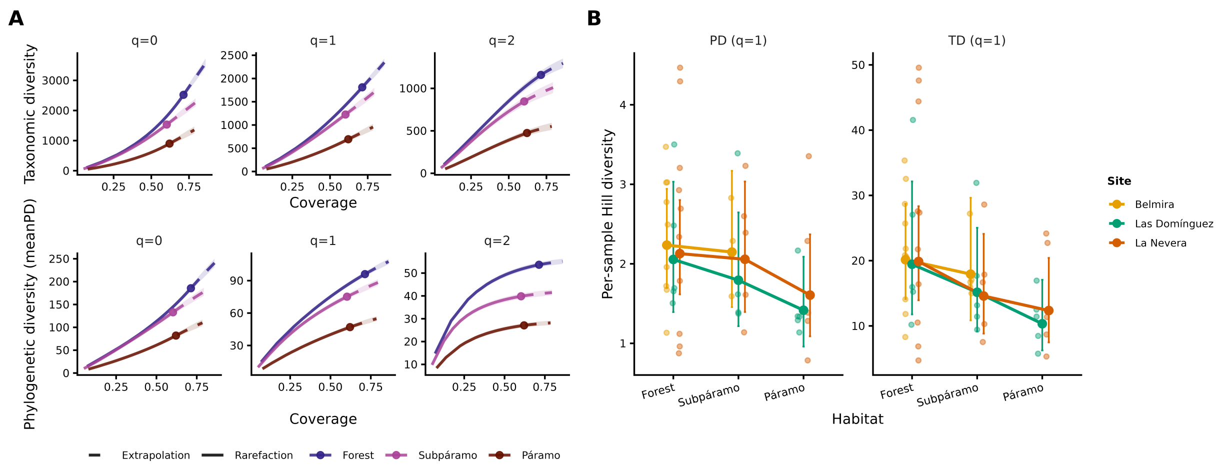

Figure 3: Diversity decline across the elevational gradient. Panel A shows iNEXT3D coverage-based rarefaction/extrapolation curves for TD and PD at $q = 0, 1, 2$ by habitat. Panel B shows per-sample Hill diversity trajectories ($q = 1$) for TD and PD, with model-estimated means ± 95% CI for BEL, DOM, and NV (MA excluded). Habitat × site interactions were non-significant for both metrics (TD $F_{4,50} = 1.52$, $p = 0.21$; PD $F_{4,50} = 1.22$, $p = 0.31$), supporting parallel diversity erosion across regions.

Section 6: Lineage-specific responses

6.1 Beta regression models (Tables S16–S18)

To quantify elevational trends in the relative abundance of dominant fungal lineages, we fitted beta-regression models to order-level relative read abundances. Models were fitted separately for each order using a logit link, with elevation as a continuous predictor and site included as a fixed effect. Relative abundances were calculated as proportions of total reads per sample and transformed using a Smithson–Verkuilen adjustment.

We focus on helotiales and sebacinales, the two most abundant orders, which also showed the only significant elevation responses. For both orders, fitted relationships and raw data are shown in Figure 4 (Panels A-B), while full model summaries are provided in Tables S16 and S17. Elevation effects for the eight most abundant orders are summarized in Table S18.

Model diagnostics were examined to assess fit and identify potential violations of distributional assumptions. Diagnostic plots include Pearson residuals versus fitted values and elevation, quantile residual Q–Q plots, and leverage values (Figures S4–S5).

Table 14: Table S16. Beta regression summary for Helotiales (mean model coefficients and precision (phi)).

Model	Term	Estimate	SE	z	p
helotiales	(Intercept)	-3.2040	1.2559	-2.5512	0.0107
helotiales	elevation	0.0010	0.0004	2.3959	0.0166
helotiales	siteDOM	-0.7291	0.3651	-1.9969	0.0458

Model	Term	Estimate	SE	z	p
helotiales	siteMA	0.0050	0.3871	0.0130	0.9896
helotiales	siteNV	-0.8864	0.3346	-2.6494	0.0081
helotiales	(phi) (phi)	6.2698	1.0711	5.8534	0.0000

Table 15: Table S17. Beta regression summary for Sebacinales (mean model coefficients and precision (phi)).

Model	Term	Estimate	SE	z	p
sebacinales	(Intercept)	0.4169	1.3941	0.2990	0.7649
sebacinales	elevation	-0.0010	0.0005	-2.1320	0.0330
sebacinales	siteDOM	0.7196	0.4162	1.7289	0.0838
sebacinales	siteMA	0.3029	0.4603	0.6580	0.5105
sebacinales	siteNV	0.9779	0.3714	2.6333	0.0085
sebacinales	(phi) (phi)	10.1810	2.0241	5.0298	0.0000

Table 16: Table S18. Elevation effects from beta regression models for the dominant orders (mean submodel), with pseudo-R^2.

order	term	estimate	se	z	p_value	pseudo_R2
helotiales	elevation	1e-03	4e-04	2.3959	0.0166	0.2133
sebacinales	elevation	-1e-03	5e-04	-2.1320	0.0330	0.2261
chaetothyriales	elevation	1e-04	4e-04	0.2327	0.8160	0.0239
sclerococcales	elevation	7e-04	5e-04	1.3758	0.1689	0.1330
agaricales	elevation	-7e-04	4e-04	-1.7597	0.0785	0.0929
pleosporales	elevation	-4e-04	4e-04	-1.0668	0.2861	0.1528
hypocreales	elevation	-2e-04	3e-04	-0.7300	0.4654	0.2397
leotiales	elevation	-2e-04	3e-04	-0.6812	0.4957	0.2964

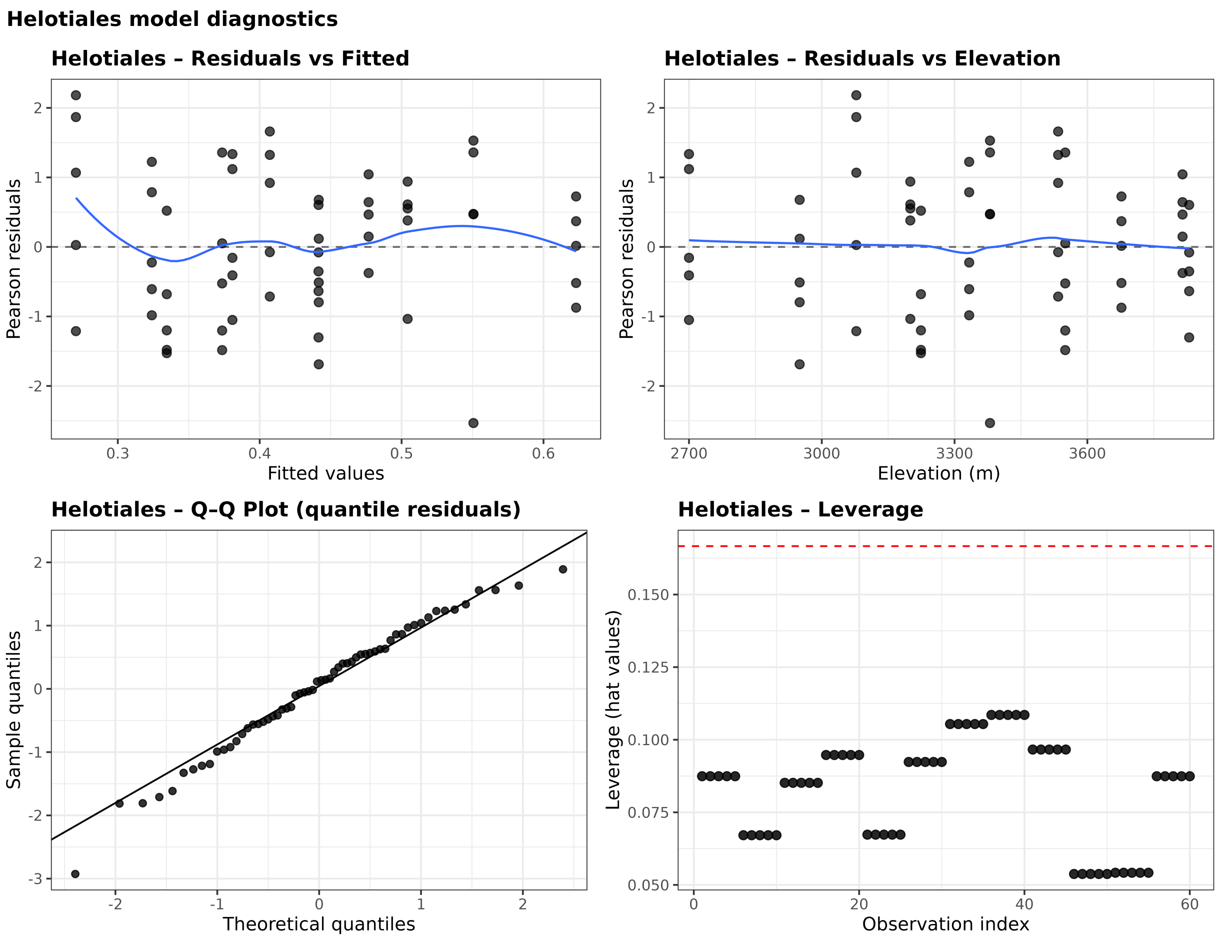

Figure S4: Diagnostic plots for the beta-regression model of Helotiales. *Code: see "Beta-regression models" below.*

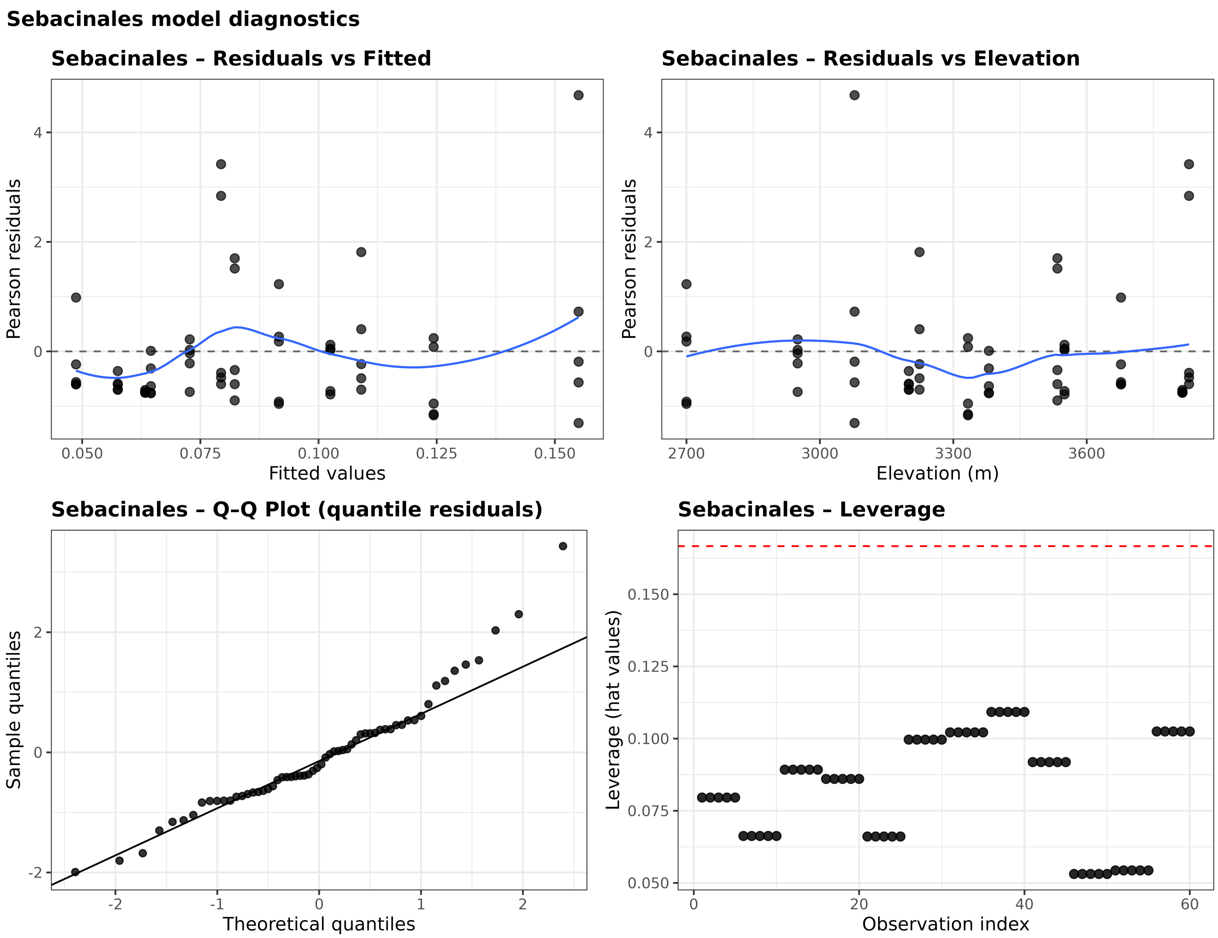

Figure S5: Diagnostic plots for the beta-regression model of Sebacinales. *Code: see "Beta-regression models" below.*

6.2 Figure 4: Elevational trends and OTU habitat responses

Three-panel composite figure combining order-level elevational trends (beta-regression for the two most abundant and responsive orders) with OTU-level habitat associations from the GLLVM model. Panel A and B show fitted beta-regression relationships for *Helotiales* and *Sebacinales* across sampling sites with raw data overlaid. Panel C presents the genus-level GLLVM heatmap showing weighted habitat contrast effects (páramo and subpáramo relative to forest) for the most responsive taxa.

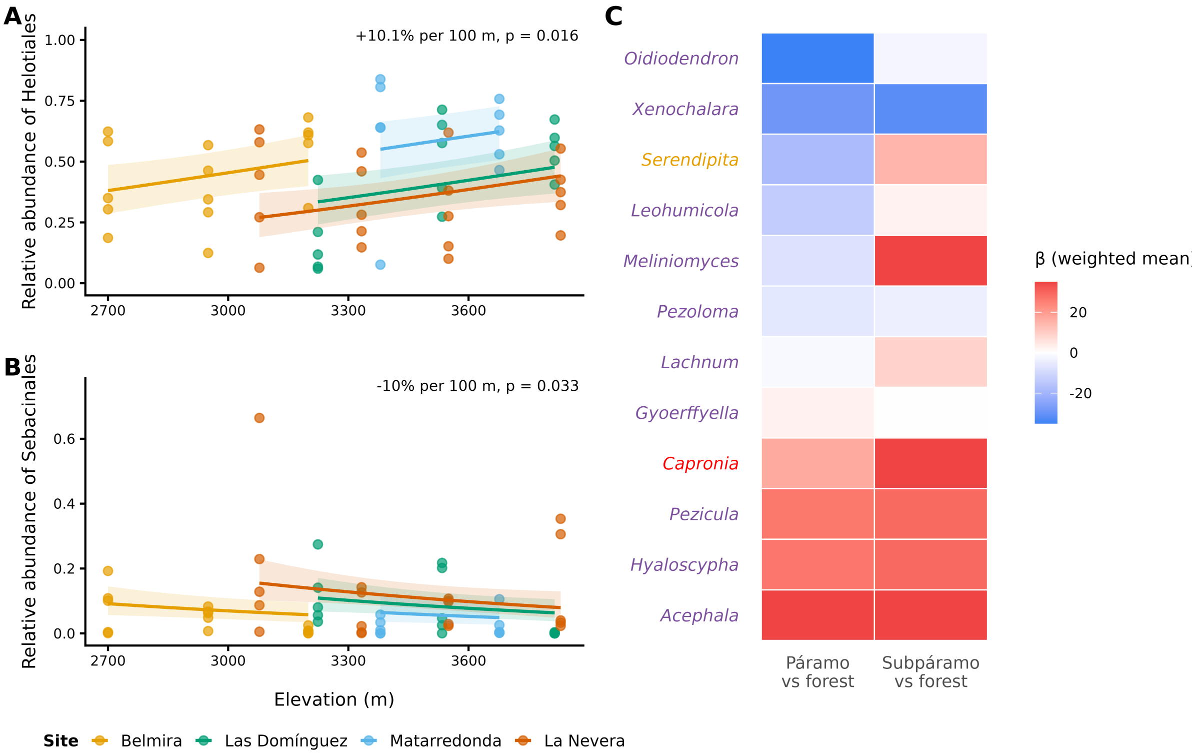

Figure 4: Elevational trends and OTU-level habitat responses. Panel A: Helotiales relative abundance versus elevation (approximately +10.1% per 100 m, p = 0.016). Panel B: Sebacinales relative abundance versus elevation (approximately -10.0% per 100 m, p = 0.033). Panel C: GLLVM heatmap of genus-level coefficients for páramo vs forest and subpáramo vs forest contrasts.

6.3 GLLVM: OTU-level habitat responses

We modelled OTU counts using a generalized linear latent variable model (GLLVM; negative binomial distribution, log link), with habitat (forest, subpáramo, páramo) as a fixed effect, log library size as an offset, and random intercepts for site and individual plants (Unique_ID). One latent variable was included to account for residual correlation among taxa. The final model (log-likelihood = −37,616; AIC = 80,295) provided the best balance between fit and parsimony. It substantially outperformed earlier versions with nested random effects (AIC = 186,832), and attempts to increase the number of latent variables to two worsened performance (ΔAIC   12,700) and produced a singular information matrix, indicating over-parameterization. Simplifying the random-effects structure and filtering rare taxa also improved convergence and reduced overdispersion while retaining ecological signal.

Model fit was evaluated using Dunn–Smyth residual diagnostics (Fig. S6). Habitat coefficients ( ) were extracted for each OTU from the fixed-effects matrix. To obtain genus-level responses, OTUs were restricted to those assigned to a genus (excluding incertae sedis) and weighted by their total read abundance across samples. For each genus, we computed the abundance-weighted mean   for páramo and subpáramo relative to forest. Uncertainty was summarized using a weighted bootstrap (R = 500) resampling OTUs within genera with probabilities proportional to total abundance; 2.5–97.5% quantiles were used as 95% confidence intervals. These summaries were used to generate the heatmap and barplot shown in Figure 4. The full OTU-level coefficient table is provided in Data_S2_GLLVM_table.csv.

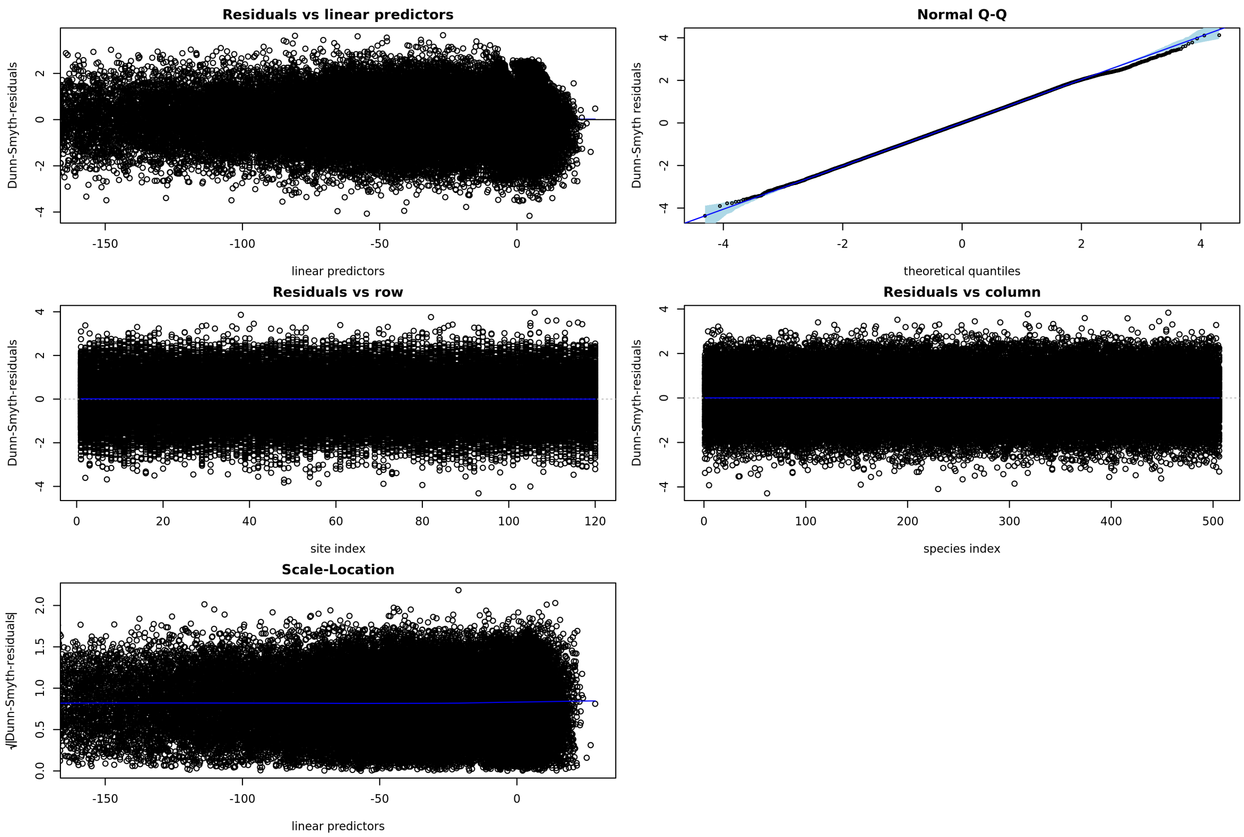

Figure S6: Diagnostic plots for the negative-binomial GLLVM (Dunn–Smyth residuals). *Code: see "GLLVM model fitting + coefficient summaries" below.*

For transparency, we provide a short preview of the genus-level GLLVM summaries below. The complete OTU- and genus-level coefficient table is available as Data S2 (Data_S2_GLLVM_table.csv).

Table 17: Data S2 (preview). Selected columns from the GLLVM genus-level summary. Column names are abbreviated here: n_OTUs = number of OTUs assigned to the genus; beta_P = w_mean_beta_paramo and beta_S = w_mean_beta_subparamo (abundance-weighted mean GLLVM coefficient for paramo and subparamo, relative to forest); enrich_P = perc_enriched_paramo_w and enrich_S = perc_enriched_subparamo_w (abundance-weighted percentage enrichment in paramo and subparamo, relative to forest). The full table with all metrics, under its original column names, is provided as a CSV file.

Order	Genus	n_OTUs	beta_P	beta_S	enrich_P	enrich_S
Coniochaetales	Coniochaeta	2	254.379	-25.536	2.048010e+115	-1.000000e+02
Helotiales	Acephala	1	78.144	69.862	8.663205e+35	2.190394e+32
Helotiales	Hyaloscypha	3	26.861	29.053	3.743955e+14	6.764441e+13
Helotiales	Pezicula	5	26.640	29.280	1.824961e+09	-9.894700e+01
Helotiales	Lachnum	3	-1.529	9.013	1.016873e+08	2.107609e+04
Sclerococcales	Sclerococcum	4	24.669	-8.648	2.210442e+06	-9.989400e+01
Mortierellales	Mortierella	5	-7.650	-34.063	1.011728e+04	-1.000000e+02
Helotiales	Gyoerffyella	3	2.739	0.188	1.021700e+01	-9.967900e+01
Ostropales	Cryptodiscus	3	-15.238	12.611	-1.947200e+01	6.019926e+04
Helotiales	Pezoloma	2	-6.125	-4.096	-9.964100e+01	-9.684900e+01

Section 7: Functional guilds

Functional guilds from root samples were assigned using FUNGuild based on the curated taxonomic strings exported from the phyloseq object. OTUs with a valid confidence ranking in the FUNGuild output were retained for downstream summaries. Guild labels were additionally collapsed into a primary-guild hierarchy (Ericoid mycorrhizal > other mycorrhizal > endophyte > pathotroph > saprotroph > other) to simplify interpretation. Figure S7 summarizes mean guild-associated read abundance per sample across habitats.

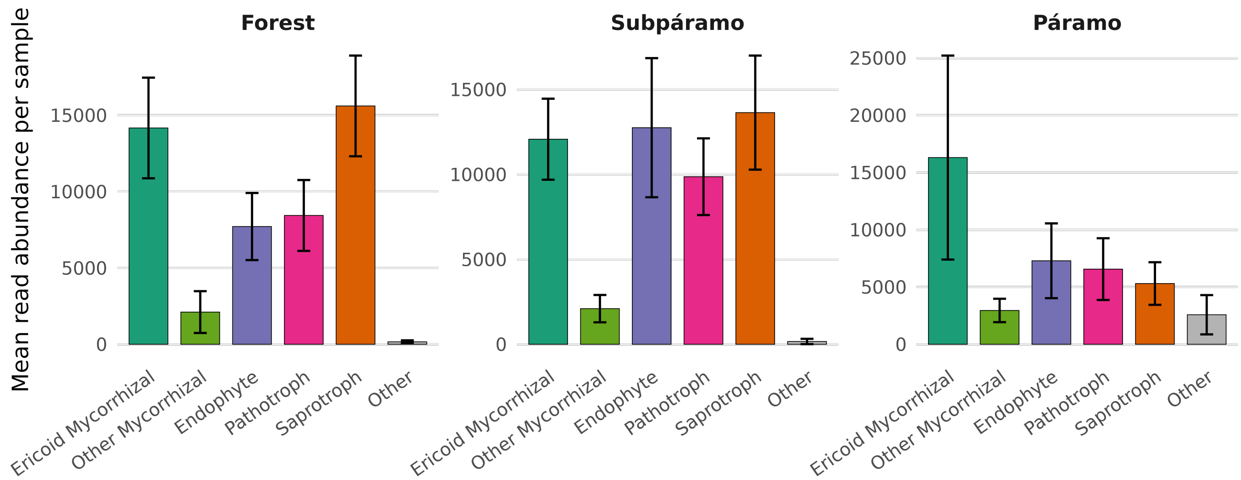

Figure S7: Mean relative read abundance per sample of fungal functional guilds across forest, subpáramo, and páramo habitats, based on FUNGuild annotations. Only root samples. *Code: see "FUNGuild functional annotation" below.*

Section 8: Phylogenetic community structure

Phylogenetic community structure was quantified using Net Relatedness Index (NRI; based on MPD) and Nearest Taxon Index (NTI; based on MNTD), calculated under a taxa-label null model with 999 randomizations using both abundance-weighted (aw) and presence–absence (pa) versions. Habitat-level summaries are reported as median (IQR) (Table S19). Habitat and site effects on abundance-weighted metrics were evaluated with linear models (habitat + site; Table S20).

Table 18: Table S19. Median (IQR) of NRI and NTI across habitats, calculated using abundance-weighted (aw) and presence–absence (pa) metrics. Positive values indicate phylogenetic clustering relative to null expectations.

Habitat	n	NRI_aw	NTI_aw	NRI_pa	NTI_pa
forest	25	0.991 (3.02)	3.053 (1.70)	0.711 (2.19)	5.764 (2.13)
paramo	15	1.435 (1.85)	3.277 (1.62)	0.831 (2.81)	4.930 (2.06)
subparamo	20	1.675 (1.77)	2.831 (1.40)	0.395 (1.50)	4.876 (2.49)

Table 19: Table S20. ANOVA summaries for linear models of abundance-weighted NTI and NRI (habitat + site).

Response	Predictor	df	F	p
NTI_aw	habitat	2	0.5106	0.6030
NTI_aw	site	3	1.8164	0.1551
NTI_aw	Residuals	54	NA	NA
NRI_aw	habitat	2	1.2729	0.2883

Response	Predictor	df	F	p
NRI_aw	site	3	3.4206	0.0235
NRI_aw	Residuals	54	NA	NA

Section 9: Soil physicochemical context

To provide environmental context for the elevational gradients sampled in this study, we quantified bulk soil physicochemical properties from one composite topsoil sample (0–15 cm) collected at each sampling location (n = 12; one per location). Because soil chemistry was measured once per location, these data are interpreted descriptively and were not used for formal statistical inference among habitats or sites.

The full soil dataset, including all measured variables is archived in the repository as “Data_S3_soil.csv”.

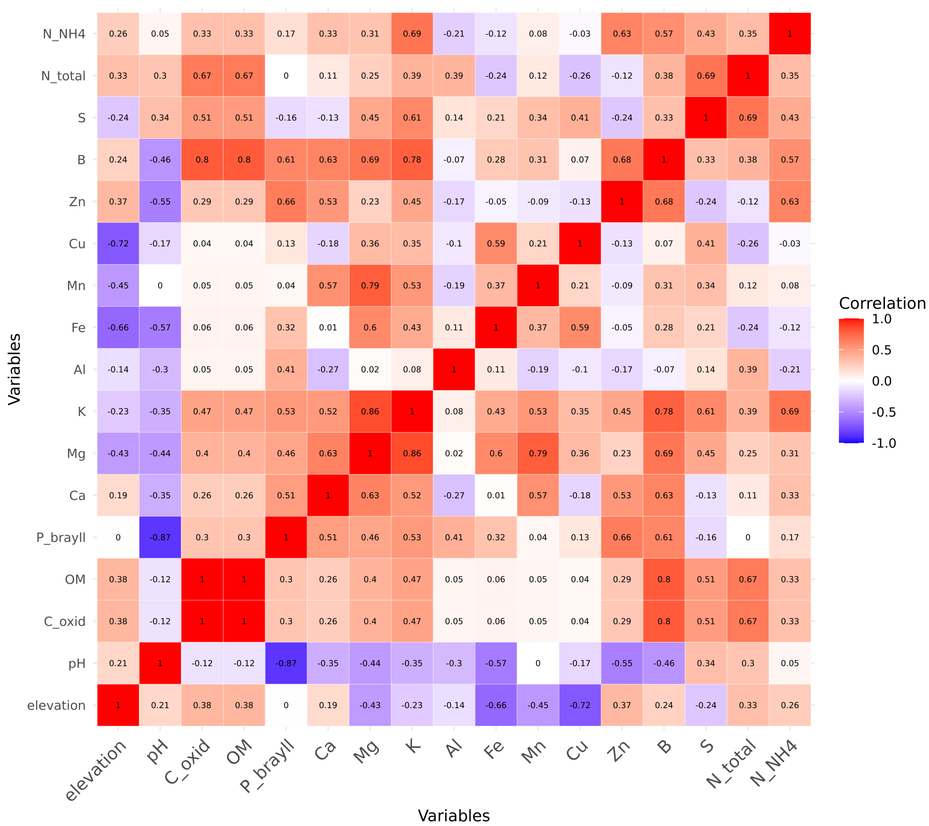

Figure S8: Correlation heatmap of bulk soil physicochemical variables measured at each sampling location (n = 12). Colors indicate Pearson correlation among soil variables and elevation.